Mid-ordinate rule

Basic skills

Q1. $a=0 \quad b=10$

a) $n=10 \quad h=\frac{10-0}{10}=1$ b) $h=2$

c) $h=2.5$

Q2. $h=2 \quad x_0=5$

a) $x_1=x_0+h=5+2=7$ b) $x_2=9$

c) $x_3=11$, d) $x_5=15$

2h

Q3. $f(x)=x^2 \quad x_{0.5}=2.5$

$y_{0.5}=f(x_{0.5})$

$=f(2.5)=2.5^2$

$=6.25$

Q4. As table has 4 values

$n=4,\ a=1,\ b=3$ so $h=0.5$

$x_{i+0.5}$ increases by 0.5

so $\int_1^3 f(x) = h\left(y_{0.5} + y_{1.5} + y_{2.5} + y_{3.5}\right)$

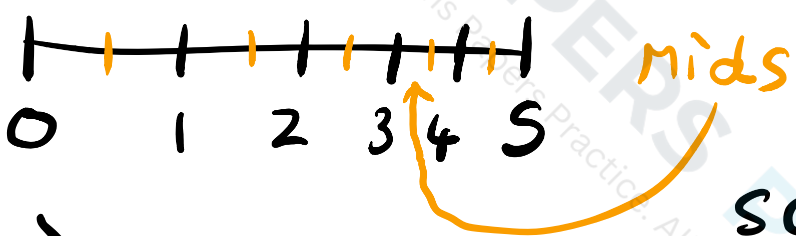
$= 0.5\left(6 + 3.5 + 2 + 1\right)$

$= 0.5 \times 12.5 = 6.25$

Competency skills

Q5. $\int_0^5 e^x \, dx$ 5 strips

so $h = \frac{5-0}{5} = 1$



so

a)

$x_{0.5} = 0.5$ $y_{0.5} = e^{0.5}$

$x_{1.5} = 1.5$ $y_{1.5} = e^{1.5}$

$x_{2.5} = 2.5$ $y_{2.5} = e^{2.5}$

$x_{3.5} = 3.5$ $y_{3.5} = e^{3.5}$

$x_{4.5} = 4.5$ $y_{4.5} = e^{4.5}$

b) $\int_0^5 e^x \, dx = 1\left(e^{0.5} + e^{1.5} + e^{2.5} + e^{3.5} + e^{4.5}\right)$

≈ 141.4 (1 d.p)

c) we can do the analytical int

$$\int_0^5 e^x \, dx = \left[e^x\right]_0^5 = e^5 - e^0$$
$$= e^5 - 1$$
$$\approx 147.4 > 141.4$$

Actual Est

so underestimate

Q6. $\int_0^5 e^{x^2} dx \quad n = 4 \quad h = \frac{5}{4} = 1.25$

$x_0 = 0 \quad x_1 = 1.25$

$x_{0.5} = 0.625$ $\quad y_{0.5} = e^{0.625^2} \approx 1.478$

$+h$

$x_{1.5} = 1.875$ $\quad y_{1.5} = e^{1.875^2} \approx 33.6$

$+h$

$x_{2.5} = 3.125$ $\quad y_{2.5} = e^{3.125^2} \approx 17424$

$+h$

$x_{3.5} = 4.375$ $\quad y_{3.5} = e^{4.375^2} \approx 205431869$

$$\int_0^5 e^{x^2} dx = 1.25(1.478 + 33.6 + 17424 + 205431869)$$

$$\approx 256811659$$

Q7. $\int_{-\pi}^{\pi} \sin^4 x \, dx$ $\quad n = 6 \quad h = \frac{\pi - -\pi}{6} = \frac{\pi}{3}$

$x_0 = -\pi \qquad x_1 = -\frac{2\pi}{3}$

$x_{0.5} = -\frac{5\pi}{6} \qquad y_{0.5} = \sin^4\left(-\frac{5\pi}{6}\right) = \frac{1}{16}$

$+h$

$x_{1.5} = -\frac{\pi}{2} \qquad y_{1.5} = 1$

$+h$

$x_{2.5} = -\frac{\pi}{6} \qquad y_{2.5} = \frac{1}{16}$

$+h$

$x_{3.5} = \frac{\pi}{6} \qquad y_{3.5} = \frac{1}{16}$

$+h$

$x_{4.5} = \frac{\pi}{2} \qquad y_{4.5} = 1$

$+h$

$x_{5.5} = \frac{5\pi}{6} \qquad y_{5.5} = \frac{1}{16}$

symmetry of $\sin^4 x$

so $\int_{-\pi}^{\pi} \sin^4 x \, dx = \frac{\pi}{3}\left(\frac{1}{16} + 1 + \frac{1}{16} + \frac{1}{16} + 1 + \frac{1}{16}\right)$

$= \frac{\pi}{3} \times 2.25 = \frac{3\pi}{4}$

could use $\int_{-\pi}^{\pi} \sin^4 x \, dx = 2\int_{0}^{\pi} \sin^4 x \, dx$

3 strips

Q8. $\int_0^5 e^x \, dx \quad n = 8 \quad h = \frac{5}{8}$

$x_0 = 0 \qquad x_1 = \frac{5}{8}$

$x_{0.5} = \frac{5}{16} \qquad y_{0.5} = e^{\frac{5}{16}} \approx 1.3668$

$x_{1.5} = \frac{15}{16} \qquad y_{1.5} = 2.5536$

$x_{2.5} = \frac{25}{16} \qquad y_{2.5} = 4.7707$

$x_{3.5} = \frac{35}{16} \qquad y_{3.5} = 8.9129$

$x_{4.5} = \frac{45}{16} \qquad y_{4.5} = 16.6515$

$x_{5.5} = \frac{55}{16} \qquad y_{5.5} = 31.1091$

$x_{6.5} = \frac{65}{16} \qquad y_{6.5} = 58.1194$

$x_{7.5} = \frac{75}{16} \qquad y_{7.5} = 108.5814$

so $\int_0^5 e^x \, dx \approx \frac{5}{8}(1.3668 + \ldots + 108.5814)$

$= 145.0$ (1 d.p) A better approximation

Advanced skills

Q9. $\int_1^3 \ln x \, dx$ Lets use 4 strips

$n = 4 \quad h = \frac{3-1}{2} = 0.5 \quad x_0 = 1 \quad x_1 = 1.5$

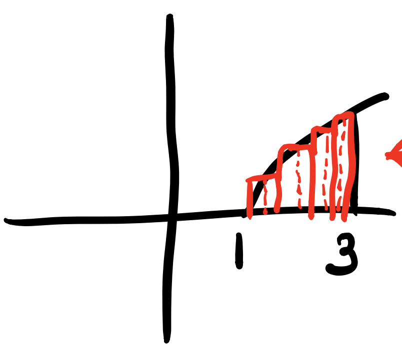
$x_{0.5} = 1.25 \quad y_{0.5} = \ln(1.25)$

$x_{1.5} = 1.75 \quad y_{1.5} = \ln(1.75)$

$x_{2.5} = 2.25 \quad y_{2.5} = \ln(2.25)$

$x_{3.5} = 2.75 \quad y_{3.5} = \ln(2.75)$

$$\int_1^3 \ln x \, dx \approx 0.5(\ln(1.25) + \ldots + \ln(2.75))$$
$$= 1.303$$

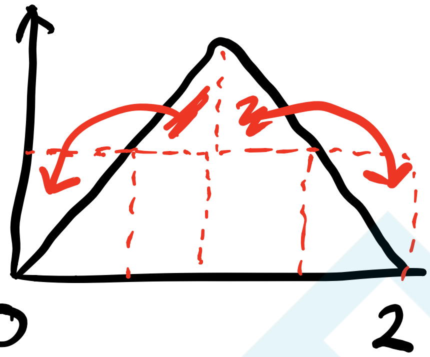


← Rectangles estimation of Area

Q10. A constant function as it is a rectangle i.e $f(x) = c \quad x \in \mathbb{R}$

Any function such that overestimate parts are equal to underestimate parts

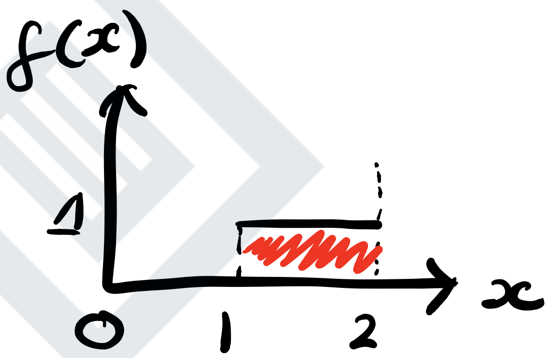
Another is



$$f(x)=\begin{cases} x & 0<x\le 1 \\ 2-x & 0<x\le 2 \end{cases}$$

Q11. $f(x)=\lfloor x \rfloor$

$\int_0^2 f(x)\,dx$ is Area so draw graph



Area is just $1\times1=1$ $\boxed{1}$

For mid-ord

$x_{0.5}=0.25$ $\quad y_{0.5}=0$ $\quad \lfloor 0.25 \rfloor = 0$ as round down

$x_{1.5}=0.75$ $\quad y_{1.5}=0$

$x_{2.5}=1.25$ $\quad y_{2.5}=1$

$x_{3.5}=1.75$ $\quad y_{3.5}=1$

Another example of exact ANS by mid

so $\int_0^2 f(x)\,dx \approx 0.5(0+0+1+1)=1$